Quadruped SLAM and exploration

Measured results and comparison with published systems | run 20261024_210638 | simulation

Test conditions

Item	Value
Platform	CoppeliaSim simulation, multi-room arena (8 rooms, about 10 x 10 m)
Robot and sensing	quadruped; forward RGB-D camera converted to a 2D scan; gait odometry; simulated gyro
Algorithm	2D scan-matching SLAM with loop closure and a 2D occupancy-grid map; CPU only
Ground truth	simulator pose of the robot body (used for scoring only)
Computer	AMD Ryzen 7 7735HS with Radeon Graphics
Run	1 run, 1800 s of simulated time, 158.0 m travelled; ended at the time limit

1. Task and exploration

Metric	Value	What it measures
Task completed	no	Whether the mission reported every reachable frontier explored
Simulated time	1800 s	Robot time of the run
Distance travelled	158.0 m	Path length of the robot from ground truth
Average speed	0.088 m/s	Distance travelled divided by simulated time
Rooms completed / found	6 / 8	Rooms marked fully explored / rooms the mapper segmented
Explored free area	84.3 m2	Floor area observed as free space
Map observed	80.1 %	Share of the 11 x 11 m map grid that is no longer unknown
Frontiers remaining	48	Boundaries between known and unknown space left at the end
Stuck events	4	Times the anti-stuck watchdog fired
Unreachable targets	9	Targets the path planner could not reach

2. Localization (position and heading, against ground truth)

Metric	Value	What it measures
ATE RMSE	31.0 cm	Root-mean-square position error over every control step (0.05 s), no alignment
ATE median / maximum	27.1 / 63.1 cm	Median and worst position error
ATE as share of path	0.20 %	ATE RMSE divided by distance travelled
Final error	38.5 cm (0.24 % of path)	Position error at the end of the run
RPE translation	9.3 cm per 1 m (7.4 %)	Error of the motion over each 1 m of true path (local accuracy)
RPE rotation	1.30 deg per m	Heading error of the motion over each 1 m of true path
Heading error mean / maximum	0.74 / 4.9 deg	Orientation error against ground truth
Loop closures	68	Revisits recognized and used to correct drift
Scan-match acceptance	26.9 %	Share of SLAM updates in which the scan match passed its checks (otherwise odometry is used)
Tracking-loss episodes	757 (548 s)	Runs of 10 or more consecutive updates without an accepted scan match

3. Mapping (against a reference grid built with the true pose)

Metric	Value	What it measures
Wall accuracy	12.9 cm	Mean distance from each mapped wall cell to the nearest reference wall cell
Wall completion	6.4 cm	Mean distance from each reference wall cell to the nearest mapped wall cell
Completion ratio (5 cm)	54.1 %	Share of reference wall cells matched within 5 cm

Chamfer / Hausdorff-95	9.6 / 30.2 cm	Mean and 95th-percentile two-way wall distance
Precision / recall / F1	66.8 / 91.3 / 77.2 %	Wall-cell agreement with the reference within 15 cm
Free-space IoU	81.2 %	Overlap of mapped free space with reference free space
Map update time (median / p95)	20.3 / 36.9 ms	Time to integrate one depth frame into the map
Map memory	0.19 MB	Size of the occupancy grid (220 x 220 cells of 5 cm)

4. Processing time and resources

Agent	Rate (Hz)	Calls	Errors	Median (ms)	p95 (ms)	Max (ms)
PerceptionAgent	10	36001	0	9.14	17.43	1986
SLAMAgent	10	36001	0	10.62	23.64	85
MappingAgent	10	36001	0	20.30	36.93	179
FrontierAgent	1	3601	0	155.00	220.89	577
PlanningAgent	5	7201	0	0.42	127.62	1061
LocomotionAgent	20	36001	0	7.25	10.72	52
SupervisorAgent	2	3601	0	0.02	0.03	0

Time of one call to each agent's step function (median, 95th percentile, maximum).

Metric	Value	What it measures
Control-loop step (median / p95 / max)	91 / 380 / 2248 ms	Wall time of one whole control step (all agents)
Real-time factor	0.32	Simulated seconds per wall-clock second (1 = real time)
Wall-clock time	5666 s	Time the computer needed for the run
Process CPU (mean / p95)	86 / 106 % of one core	CPU use of the Python process; simulator excluded
Memory (mean / peak)	1037 / 2973 MB	Resident memory of the Python process
GPU	not used	No part of the pipeline runs on the GPU

5. Component ablation (offline replay of the recorded run)

The same recorded sensor data is replayed with one component switched off; the live robot is not changed.

Configuration	ATE RMSE (cm)	Final error (cm)	Heading error (deg)	Loop closures
full system	31.0	38.5	0.74	68
Without scan matching (odometry + gyro only)	264.8	135.7	44.62	0
Without loop closure	54.4	90.1	0.71	0
Without degeneracy filter	32.9	30.7	0.74	70
Without Manhattan heading prior	168.9	215.9	18.78	71

Comparison with published systems

Published values are quoted from the cited papers and product pages; 'derived' means two published numbers were divided. Test scenes, sensors, ground truth and map representations differ from ours as listed in the Test conditions column.

6. SLAM as a whole: pose accuracy, map accuracy and cost

System	Test conditions	Pose: ATE RMSE (cm)	Map: accuracy / completion (cm) / completion ratio (%)	Tracking per frame (ms)	Mapping per frame (s)	Map memory (MB)
Ours	simulated 8-room arena, 158 m; forward RGB-D to 2D scan + gait odometry + gyro; 2D occupancy grid; CPU [this work]	31.0	12.9 / 6.4 / 54.1	10.6	0.020	0.19
MNE-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.398	1.681 / 1.509 / 97.0	22.1	0.14	24.45
Loopy-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.369	1.593 / 2.645 / 91.1	-	-	-

GO-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.404	2.091 / 2.415 / 93.2	-	-	-
ESLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.657	2.082 / 1.754 / 96.0	68.5	0.29	20.73
Co-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	1.059	2.104 / 2.082 / 93.4	54.8	0.15	26.39
Point-SLAM	Replica scenes, RGB-D, neural 3D map, GPU [1]	0.559	1.935 / 2.779 / 90.5	29.7	10.47	31.42

7. Pose accuracy in multi-room scenes

System	Test conditions	ATE RMSE (cm)
Ours	own 8-room arena, 158 m [this work]	31.0
MNE-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.02 / 0.59 / 0.48
ESLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.17 / 0.79 / 0.76
Loopy-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	1.43 / 0.60 / 0.64
ORB-SLAM3	Replica multi-room apartments 1 / 2 / 0 [1]	4.93 / 1.36 / 1.07
Swarm-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.56 / 5.61 / 1.80
CP-SLAM	Replica multi-room apartments 1 / 2 / 0 [1]	5.73 / 1.85 / 0.91

8. Pose accuracy on legged robots

System	Test conditions	Path (m)	ATE RMSE (cm)	ATE / path (%, derived)	CPU (%) / RAM (GB)
Ours	simulation, simulator pose [this work]	158	31.0	0.20	86 / 1.0
A1 SLAM	real Unitree A1, lab, motion capture [3]	5.0-5.3	4.3-14.5	0.8-2.9	-
Cartographer	real Unitree A1, lab, motion capture [3]	5.0-5.3	4.0-139.7	0.8-27.9	-
FAST-LIVO2, LiDAR + camera + IMU (Honeywell)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	10.5	0.04	125 / 3.91
ORB-SLAM3, stereo (global shutter)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	33.0	0.14	290 / 1.16
RTAB-Map, stereo (global shutter)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	57.9	0.25	116 / 0.75
RTAB-Map, RGB-D (ZED2i)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	241.3 (successful runs only)	1.03	-
ORB-SLAM3, RGB-D (ZED2i)	real ANYmal D, mostly outdoor, 7 missions [4]	145-332	821.4	3.50	-

9. Processing time per update

System	Quantity	Value	Hardware / note
Ours	SLAM update, median	10.6 ms	CPU only, AMD Ryzen 7 7735HS with Radeon Graphics [this work]
ORB-SLAM2 (RGB-D)	tracking per frame	25.58 ms	Intel Core i7-4790, TUM fr3_office [2]
MNE-SLAM	tracking per frame	22.1 ms	authors' hardware [1]
A1 SLAM / Cartographer	optimization per scan	1.65 / 0.96 ms	optimizer call only [3]

10. Exploration speed

System	Test conditions	Distance (m)	Time (s)	Average speed (m/s)
Ours	quadruped, forward RGB-D, 10 x 10 m arena [this work]	158	1800 (not completed)	0.088

DSVP	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1511	931	1.62 (derived)
TARE	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1209	658	1.84 (derived)
ARiADNE	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1053	610	1.73 (derived)
HEADER	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1030	576	1.79 (derived)
FARE (LLM + RL)	wheeled robot, 16-channel 3D LiDAR, Gazebo indoor, max 2 m/s [5]	1048	590	1.78 (derived)
ANYmal (product page)	quadruped, 360 deg LiDAR + depth cameras [7]	-	-	0.75 normal walking
Spot (product page)	quadruped, 360 deg terrain sensing [6]	-	-	1.6 maximum
Go2 (product page)	quadruped, 4D LiDAR [8]	-	-	about 5 maximum (lab)

Map accuracy / completion / completion ratio: mean wall-to-wall distances and share of wall cells within 5 cm. Ours are measured on 2D grid wall cells, published values on 3D reconstructed surfaces. ATE: absolute trajectory error (RMSE). Path: distance travelled in the test.

References

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